Acknowledgements

Every system of knowledge is the product not of one mind but of the lives that shaped it. The understanding that became PQNK was grown in conditions that no institution could have provided: conditions of family, friendship, adversity, and loyalty that left their mark on both the farmer and the farming.

Mian Khair Mohammad, my grandfather, was a man who understood, long before it was fashionable to say so, that knowledge is the only inheritance that cannot be taken away. In 1917, in a village fifty-six miles from the nearest school, he established a high school and opened its doors free of charge to every child who wished to learn. The institution he built changed the fate of an entire district: Toba Tek Singh went on to attain the highest literacy rate in rural subcontinental India. He left me two lessons that have governed every decision of my working life. The first: respect everything in creation, living and dead, for every form of life carries a purpose you may not yet understand. The second: theft is a cause of injustice, but it is never the thief who is the problem; punish those who create the conditions that make theft necessary. These were not convenient principles. They have cost me something at every turn. They have also been correct at every turn.

Choudhary Mohammad Sharif, my father, gave me something that parents rarely manage and that no institution can manufacture: the freedom to think, and the platform on which to test my thinking. A man of practical generosity, he understood that experimentation requires the latitude to fail without consequence. That latitude was his gift to me.

Sharifian Begum, my mother, was the foundation beneath everything. Her love was not conditional on success, not retracted in difficulty, and not diminished by the years of disruption that my work brought upon our family. She was a woman of quiet nobility who extended her support not only to her own children but to everyone who came within the orbit of her life. I carry her example in every interaction I have with the farmers who come to us.

Khalid Sharif Choudhary, my elder brother, was a qualified irrigation management professional whose practical intelligence shaped my early understanding of how water, land, and agriculture actually work, as opposed to how the textbooks say they do. The breadth of initiatives he advanced for the relevance of the agricultural sector remains underacknowledged by those who benefited most from them. He was my first real teacher, and his teaching has never left me.

Shama Asif, my wife, has lived alongside a life that most people would not have chosen. She stood with me through the experiments that did not work, the years of disruption, the professional adversity, and the long, demanding process of building something that the world was not yet ready to receive. Without her steadiness and her support, the work that this book represents would not have been possible. I am grateful to her in ways that acknowledgements pages are not adequate to express.

Naseer Ahmad, my adopted son, gave something that is perhaps the rarest of human qualities: he stood by me without condition, without calculation, and without expectation of return. Through the most difficult years of my life, he devoted himself to the well-being of the whole family with a loyalty that had nothing to do with obligation and everything to do with character. His life has been, in the truest sense, a gift to us all.

My family as a whole bore the costs of the life I chose: the setbacks, the disruptions, the unpredictable rhythms of a working life driven by conviction rather than comfort. They withstood what would have broken lesser families, and each of them went on to build lives of which I am genuinely proud. Their capacity to support one another and maintain the unity of the family through every difficulty is, to me, the achievement I value most.

Among the friends whose support made me an achiever rather than merely a believer, I want to name several in particular: Dr. Zafar Altaf, Mohammad Azhar Khan, Mohammad Muslim, Hafizullah Ishaq, and Choudhary Mohammad Sidique. Each of them gave generously: their time, their networks, their confidence in what we were attempting, at moments when such support was not easy to extend and not taken for granted by those who received it.

Shaukat Javid is one of those rare individuals whose public life and private character are indistinguishable from one another: what he gives in the open is exactly what he gives behind closed doors, and neither is calculated for effect. Across the years of our friendship, I have watched him devote himself to the service of others with a consistency that never wavered according to whether anyone was watching. His work in the public domain has touched lives that he will never be fully credited for touching: communities quietly lifted, strangers helped at moments of real need, causes advanced without his name ever attached to them. That is not modesty as a strategy. It is simply how he moves through the world. I wrote to him recently, after watching him speak in a television interview, and told him what I had observed across the full span of our friendship: that the world is genuinely better because of him, and that I am proud to stand by his side. I meant every word of it then. I mean every word of it here.

Navid Rasool Mirza stands apart in this acknowledgement, and I want to say clearly why. Through the years when circumstances outside my control placed the work under its greatest strain, it was Navid who stood beside it. A prominent lawyer of both skill and principle, he gave his time, his counsel, and his protection without charge, without condition, and without any expectation of return. The debt this created is one I have never been able to settle, and I have made my peace with the fact that I never will. What I can say is this: the work contained in this book would not have survived its most difficult years without his protection, and there is no acknowledgement adequate to that. Some things can only be recorded, not repaid. This is one of them.

Ahmed Riaz Sheikh, his wife Shabina, a very loving soul, and his family occupy a place in our lives that goes beyond anything that professional acknowledgement can capture. At every moment when a door needed opening, a connection needed making, or help needed to arrive from a direction I had not anticipated, Riaz was there: quietly, reliably, and without ever making his assistance feel like a favor. His support has been unconditional in the truest sense of the word: extended without calculation, sustained without expectation, and offered with the ease that only genuine friendship produces. His family and ours are bound by ties that have grown through years of shared difficulty and shared joy. We are grateful, all of us, in ways that belong to the private record as much as the public one.

The field results that form the evidential core of this book were produced in collaboration with the farmers who trusted the system before it was proven and who committed their land to the process of proving it. Among the PQNK colleagues and early adopters to whom this work owes its practical foundation, I want to acknowledge in particular: Sultan Bhatti, Rashid Dogar, and Mubasher Riaz, and the many other farmers who gave their fields as

working laboratories and whose careful observation under real production conditions produced the documentation on which every claim in this book rests.

Syed Nayyar Uddin Ahmad, writer, cricket commentator, and a voice followed by a wide and engaged readership, gave PQNK something that technical documentation alone cannot provide: reach. A man of letters who moves with equal ease between the world of ideas and the world of public affairs, he embraced the PQNK cause with a wholehearted enthusiasm that owed nothing to obligation and everything to genuine conviction. Through his articles and through the personal advocacy he carried into his extensive circles of friends and followers, he brought the system to the attention of people who might never otherwise have encountered it. In a country where public discourse shapes what is considered possible, that kind of endorsement from a trusted and widely read voice carries real weight.

Dr. Ikramul Haq and Huzaima Bukhari represent a rare thing in public intellectual life: a partnership in which two formidable minds, each distinguished in their own right, bring their combined authority to bear on the questions that matter most. Nationally and internationally recognized as economists whose writings command serious attention across government, academia, and the broader reading public, they have consistently used that platform to argue for what the agro-economic evidence demands: that the transformation of Pakistan's food production system is not a peripheral concern but a civilizational one. Their support for PQNK has been woven into their articles on agricultural economy and extended through every relevant circle in which they move and are heard. To have one's work endorsed by economists of their standing is not merely an honour; it is a signal to policymakers and institutions that the case for PQNK rests on ground that goes well beyond the field, that it is, at its foundation, also a case about national solvency, food sovereignty, and the long-term terms on which this country feeds itself. I am deeply grateful for their commitment to that argument and for the generosity with which they have made it their own.

Choudhary Arshad Dad, former Secretary General of Pakistan Tehreek-e-Insaf, occupies a different but equally important place in this acknowledgement. The work of developing a sound system of production is one thing; the work of carrying that system into the rooms where policy is made is another, and it is work that requires a particular kind of person: someone who understands both the science and the politics, and who is willing to advocate without personal agenda. Choudhary Arshad Dad has done exactly that. His efforts to bring PQNK to the attention of political and governmental circles have been sustained, selfless, and driven entirely by conviction in what the system represents for Pakistan's agricultural future. That kind of advocacy, unglamorous, uncompensated, and often thankless, is what determines whether a system of knowledge remains in the field or reaches the scale it deserves.

From Lahore University of Management Sciences, the intellectual engagement of Dr. Abubakr Muhammad and Dr. Muhammad Tariq brought rigor and institutional credibility to work that the formal agricultural research establishment had not yet found the frameworks to evaluate. Their support is appreciated.

Two international figures played a distinctive role in bringing PQNK to the attention of the wider world, and their contributions deserve acknowledgement of a particular kind.

Professor Dr. Norman Uphoff of Cornell University, the scholar behind the System of Rice Intensification and one of the most consequential voices in the global conversation on

ecological agriculture, supported this work with a generosity of spirit that went well beyond professional courtesy. He engaged with the PQNK findings seriously and at length, reviewed and refined the articles through which PQNK's results were first communicated to an international audience, and lent his standing to work that was, at the time, outside the established frameworks of agricultural science. His intellectual endorsement opened doors that would otherwise have remained closed, and his encouragement at critical moments gave the work a confidence it needed. I am deeply grateful.

Mark Fulford, a dedicated farmer and farmers' adviser from Maine, USA, brought something equally valuable: the perspective of a practitioner who understood from his own ground-level experience exactly what PQNK was claiming and exactly why it mattered. His encouragement was not the encouragement of someone who had read the papers; it was the encouragement of someone who recognized the truth in the field results because he had been looking for that truth himself. His sustained support over the years of PQNK's development contributed more than he perhaps knows.

Marianne Julia Wortel, a Dutch national who came to Pakistan in 2021 and chose to make it her permanent home, brought something that I did not fully anticipate and have never taken for granted: a quality of independent thinking that consistently pushed the work further than it might otherwise have gone. Her contribution to PQNK has been far more than administrative. In the years of our collaboration, she challenged me to think differently about health: to take responsibility for one's own well-being rather than deferring entirely to medical convention, and I am honest enough to say that without that shift, the sustained pace of work this book required would not have been possible. She helped me see more clearly what lies behind the agrochemical system: not merely bad science or commercial interest, but a deeper structure of dependency that PQNK exists to dismantle. At moments when important decisions about PQNK's direction had to be made, her counsel was direct and sometimes uncomfortable: she stepped back from a potential engagement with the World Bank at a point when I was not yet ready to do the same, and later I understood exactly why she had. These were not peripheral contributions. They were conversations that shaped the course of the work. In the past year, as she has devoted her energies to establishing her own farm, the most honest proof of conviction a person can offer, her day-to-day role has settled into administering the seventeen WhatsApp groups through which PQNK knowledge reaches farmers across the country and beyond. She does this with the same care and daily commitment she has brought to everything. I am grateful to her for all of it.

Ahmad Masood has devoted himself to ensuring that fifty years of accumulated field knowledge does not remain inaccessible to those who need it most. His work in systematically compiling and organizing the PQNK body of knowledge, making it navigable for practitioners, students, and researchers, is unglamorous and essential. He is also working alongside farmers to establish a farmer-owned PQNK Crop Production Management Company, designed to organize the production of high-quality food and connect it directly with consumers: a practical expression of everything this system stands for.

Anwar ul Islam manages the PQNK website and ensures that the knowledge papers through which the system's findings are communicated remain current, properly catalogued, and publicly available. His contribution, given entirely as a volunteer, keeps the work in circulation long after any single conversation or field visit has ended.

Farooq Tariq has devoted his life to a proposition that the powerful have always found inconvenient: that the rights of workers, peasants, and the landless are not a charitable concern but a matter of justice, and that any serious conversation about poverty must begin with the conditions under which food is grown and by whom. As an activist whose personal commitment to human rights, food sovereignty, and the dignity of agricultural labor has shaped his life's work, he recognized in PQNK something that went beyond agronomy: a practical pathway to poverty eradication and a foundation for rebuilding the national economy from the ground up. He and his organization have championed the system in the circles where its message is most urgently needed, advocating PQNK not as a technical intervention but as a transformative social one. More concretely, he organized training programs that brought PQNK's methods directly to farmers, the people for whom the difference between a system that works and one that extracts is not an academic question but a question of survival. His support has been given with the same unflinching conviction that characterises everything he does, and it has reached communities that no conference or journal paper ever could.

Vasudha Sardar, a farmer from Maharashtra, India, whom I have the privilege of calling my younger sister, gave this book its final form. The process of transforming fifty years of field knowledge, written practice, and structured argument into a manuscript that could do justice to the system it describes required someone who combined genuine intellectual engagement with an unusual degree of patience and precision. Vasudha brought all of these qualities, along with an enormous investment of time and disciplined effort, to a task that was never simple and that she never treated as anything less than important. Without her, this book would exist in principle. Because of Behn Vasudha, it exists in print.

Asif Sharif, Lahore, 2026